

● MEDIUM

● GEOMETRY AND TRIGONOMETRY

Geometry and Trigonometry

07

10 questions · ~15 min

Q1

GEOMETRY AND TRIGONOMETRY

A cylindrical can containing pieces of fruit is filled to the top with syrup before being sealed. The base of the can has an area of 75 cm^2 , and the height of the can is 10 cm. If 110 cm^3 of syrup is needed to fill the can to the top, which of the following is closest to the total volume of the pieces of fruit in the can?

- A. 7.5 cm^3
- B. 185 cm^3
- C. 640 cm^3
- D. 750 cm^3

Q2

GRID-IN

GEOMETRY AND TRIGONOMETRY

Triangles ABC and DEF are congruent, where A corresponds to D , and B and E are right angles. The measure of angle A is 27° . What is the measure, in degrees, of angle F ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3

GRID-IN

GEOMETRY AND TRIGONOMETRY

A circle has a circumference of 31π centimeters. What is the diameter, in centimeters, of the circle?

STUDENT-PRODUCED RESPONSE

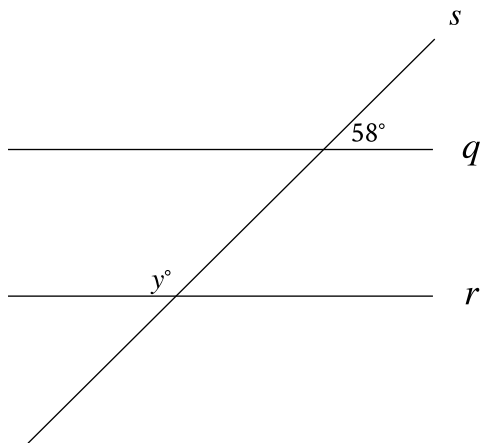
.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4

GRID-IN

GEOMETRY AND TRIGONOMETRY



Note: Figure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled s , q , and r .
- Line s intersects both line q and line r .
- At the intersection of line s and line q , 1 angle is labeled clockwise from top left as follows:
 - Top right: 58°
- At the intersection of line s and line r , 1 angle is labeled clockwise from top left as follows:
 - Top left: y°
- A note indicates the figure is not drawn to scale.

In the figure, line q is parallel to line r , and both lines are intersected by line s . If $y = 2x + 8$, what is the value of x ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3

4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5

GRID-IN

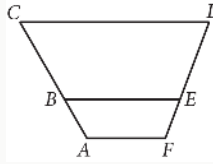
GEOMETRY AND TRIGONOMETRY

The length of each edge of a box is 29 inches. Each side of the box is in the shape of a square. The box does not have a lid. What is the exterior surface area, in square inches, of this box without a lid?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



In the figure above, $\overline{AF}$, $\overline{BE}$, and $\overline{CD}$ are parallel. Points B and E lie on $\overline{AC}$ and $\overline{FD}$, respectively. If $AB = 9$, $BC = 18.5$, and $FE = 8.5$, what is the length of $\overline{ED}$, to the nearest tenth?

- A. 16.8
- B. 17.5
- C. 18.4
- D. 19.6

Q7

GRID-IN

GEOMETRY AND TRIGONOMETRY

A circle has a radius of 2.1 inches. The area of the circle is $b\pi$ square inches, where b is a constant. What is the value of b ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Triangle ABC is similar to triangle XYZ , such that A , B , and C correspond to X , Y , and Z respectively. The length of each side of triangle XYZ is 2 times the length of its corresponding side in triangle ABC . The measure of side AB is 16. What is the measure of side XY ?

- A. 14
- B. 16
- C. 18
- D. 32

Q9

GRID-IN

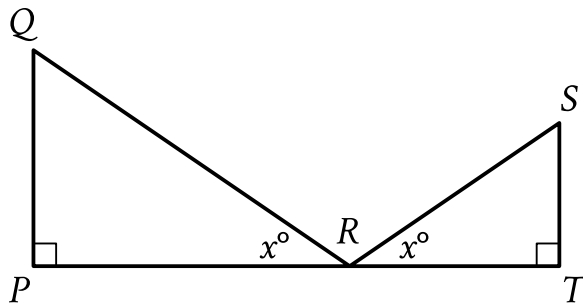
GEOMETRY AND TRIGONOMETRY

A right circular cylinder has a volume of 377 cubic centimeters. The area of the base of the cylinder is 13 square centimeters. What is the height, in centimeters, of the cylinder?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



Note: Figure not drawn to scale.

- The 2 triangles share common vertex upper R.
 - Common vertex upper R lies on line segment upper P upper T.
- Triangle upper Q upper P upper R is labeled as follows:
 - Angle upper Q upper P upper R is a right angle.
 - The measure of angle upper Q upper R upper P is x° .
- Triangle upper S upper T upper R is labeled as follows:
 - Angle upper S upper T upper R is a right angle.
 - The measure of angle upper S upper R upper T is x° .
- A note indicates the figure is not drawn to scale.

$\triangle QPR$ is similar to $\triangle STR$. The lengths represented by $\bar{ST}$, $\bar{QP}$, $\bar{PR}$, and $\bar{QR}$ in the figure are 14, 15, 20, and 25, respectively. What is the length of $\bar{SR}$?

- A. $\frac{350}{15}$
- B. $\frac{350}{20}$
- C. $\frac{210}{20}$
- D. $\frac{210}{25}$